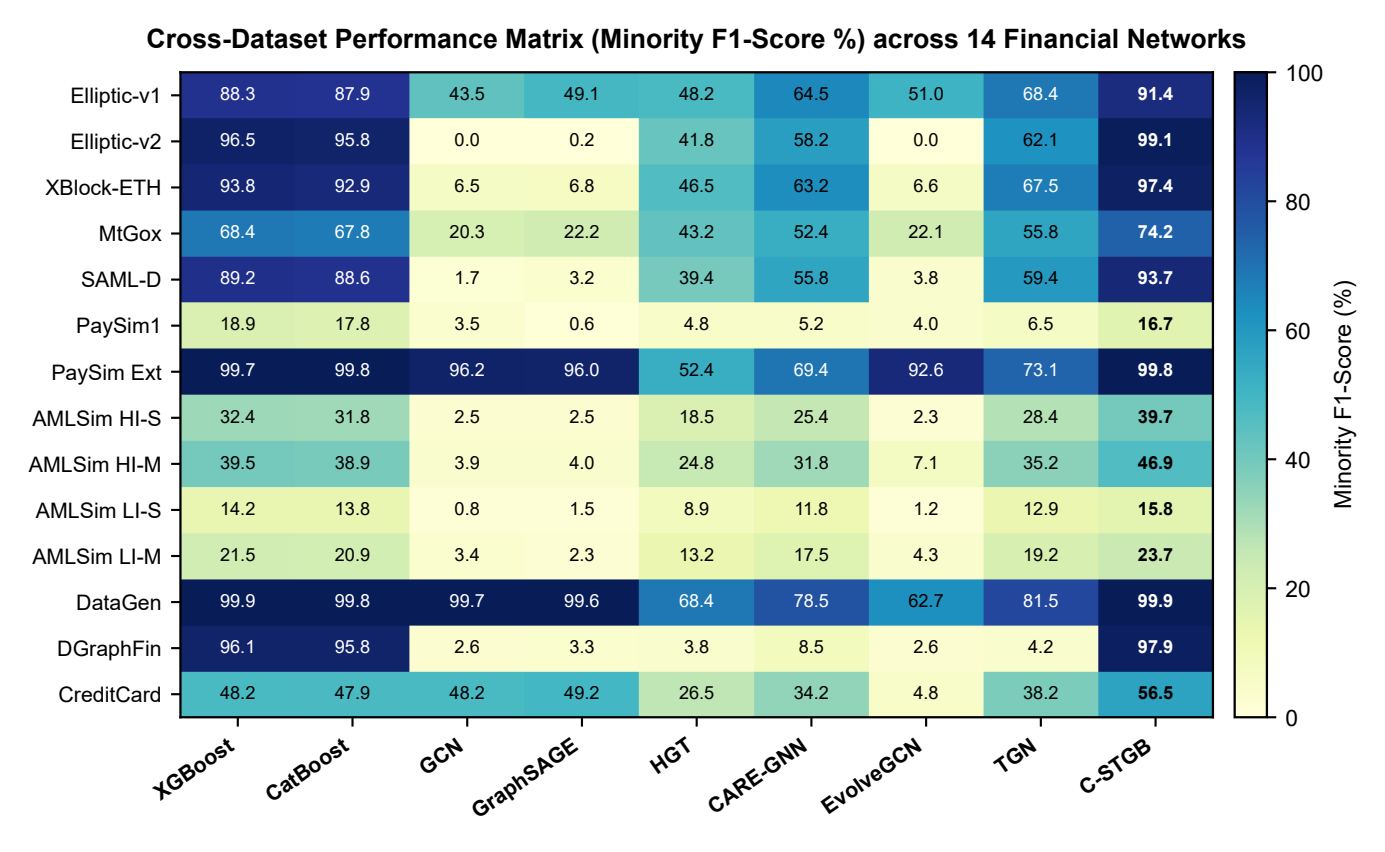

Cross-Dataset Performance Matrix (Minority F1-Score %) across 14 Financial Networks
Elliptic-v1 88.3 87.9 43.5 49.1 48.2 64.5 51.0 68.4 91.4
Elliptic-v2 96.5 95.8 0.0 0.2 41.8 58.2 0.0 62.1 99.1
XBlock-ETH 93.8 92.9 6.5 6.8 46.5 63.2 6.6 67.5 97.4
MtGox 68.4 67.8 20.3 22.2 43.2 52.4 22.1 55.8 74.2
SAML-D 89.2 88.6 1.7 3.2 39.4 55.8 3.8 59.4 93.7
PaySim1 18.9 17.8 3.5 0.6 4.8 5.2 4.0 6.5 16.7
PaySim Ext 99.7 99.8 96.2 96.0 52.4 69.4 92.6 73.1 99.8
AMLSim HI-S 32.4 31.8 2.5 2.5 18.5 25.4 2.3 28.4 39.7
AMLSim HI-M 39.5 38.9 3.9 4.0 24.8 31.8 7.1 35.2 46.9
AMLSim LI-S 14.2 13.8 0.8 1.5 8.9 11.8 1.2 12.9 15.8
AMLSim LI-M 21.5 20.9 3.4 2.3 13.2 17.5 4.3 19.2 23.7
DataGen 99.9 99.8 99.7 99.6 68.4 78.5 62.7 81.5 99.9
DGraphFin 96.1 95.8 2.6 3.3 3.8 8.5 2.6 4.2 97.9
CreditCard 48.2 47.9 48.2 49.2 26.5 34.2 4.8 38.2 56.5
XGBoost
CatBoost
GCN
GraphSAGE
HGT
CARE-GNN
EvolveGCN
TGN
C-STGB
Minority F1-Score (%)
100
80
60
40
20
0